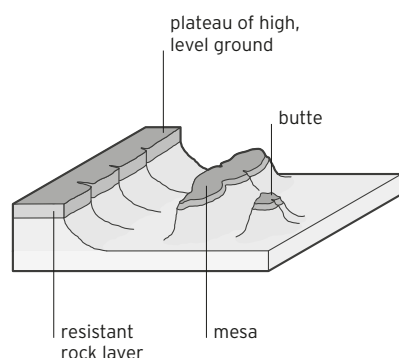


PILLOW LAVA Molten rock that is extruded onto the ocean floor and cools quickly to form pillow-shaped lumps, usually of basalt. See also *lava*.

PINGO A periglacial landform comprising a dome-shaped hillock with a core of ice. Pingos form where groundwater—often from an old lake—rises into an area where it freezes and expands, pushing up the sediments above it. See also *periglaciation*.

PLATE BOUNDARY See *boundary* (panel).

PLATEAU ▼ A large area of flat or gently undulating land whose surface is significantly higher than the surrounding landscape.



PLAYA A flat-bottomed depression in a desert basin. Seasonally, a playa may be inundated to a shallow depth, but with no natural outlet, all the water evaporates.

PLINIAN ERUPTION The largest and most violent type of volcanic eruption, involving the release of enormous quantities of ash, lava, and gas. These eruptions are so violent that they can destroy large sections of the volcano itself. See also *eruption* (panel).

PLUG A mass of igneous rock filling the vent of an active volcano. If gas or magma is trapped under the plug, pressure will build beneath it, increasing the

likelihood of the next eruption being particularly violent. See also *magma*.

PLUTON A large body of magma that solidified slowly underground to form a coarse-grained igneous rock, such as granite. See also *batholith*.

PLUTONIC Relating to igneous processes taking place, or igneous rocks formed, deep below Earth's surface. See also *batholith*, *igneous intrusion* (panel), *pluton*.

POINT BAR The area of sand, silt, or gravel on the inside of a river meander, where water flow is slowest. See also *bar*, *meander*.

POLDER An area of low-lying land reclaimed from the sea and enclosed within dikes.

POLYNA A stretch of open water in an expanse of sea-ice. See also *pack ice*, *sea-ice*.

POLYTHERMAL GLACIER A glacier whose base is partly "warm ice" (at 32°F, 0°C) and partly cold ice (below 32°F, 0°C). Typically, the central areas are warmer than the margins.

PRECIPITATE A substance that has come out of solution to form a solid deposit.

PRECIPITATION Water originating from the atmosphere that reaches Earth's surface, including rain, snow, hail, and drizzle.

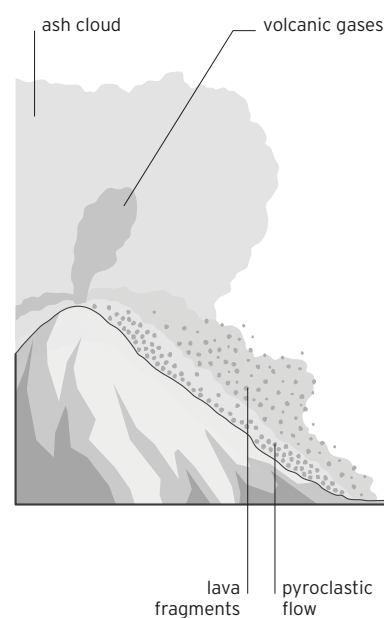
PRESSURE GRADIENT The rate of change of atmospheric pressure with distance.

PRESSURE SYSTEM A weather pattern in which air circulates around an area of low pressure (a depression, or cyclone) or high pressure (an anticyclone). See also *anticyclone*, *cyclone*, *isobar*.

PREVAILING WIND The most common direction of airflow at a particular location.

PUMICE A light, porous volcanic rock, which solidified rapidly from gas-rich frothy magma. The numerous pores, or vesicles, were gas bubbles.

PYROCLASTIC DEPOSIT An accumulation of fragments of rock, including ash and volcanic bombs, erupted from a volcano.



PYROCLASTIC FLOW ▲ A dense, fast-moving, and destructive cloud of hot ash, fragments of lava, and gas ejected violently from a volcano. See also *eruption* (panel).

R

RADIATION FOG The fog created by the night-time cooling of air close to the ground until its content of water vapor condenses. See also *fog*.

RAIN FOREST A forest with high rainfall and humidity all year round. Most

rain forests are in the tropics or subtropics, with some in temperate regions.

RAIN SHADOW The area of low rainfall downwind of a mountain range. This is produced as a result of the air shedding its moisture as it passes over the mountains.

RAINBOW A prismlike light effect caused by the Sun's rays being refracted into their constituent colors as they pass through raindrops.

RAPIDS Sections of a watercourse with a steep riverbed gradient, causing an increase in water velocity and turbulence.

RECUMBENT FOLD A fold of stratified rocks where the two limbs are almost parallel and close to the horizontal. See also *fold* (panel).

REGIONAL METAMORPHISM The processes of extreme heating and pressure by which the mineralogy and texture of rocks are transformed over a large area. See also *contact metamorphism*, *dynamic metamorphism*, *thermal metamorphism*.

REGOLITH The layer of soil and broken rock covering solid rock at Earth's surface.

RELATIVE HUMIDITY The amount of water vapor in a "parcel" of air, relative to the total amount that parcel can hold at that particular temperature. Warm air can hold more water vapor than cold air. See also *condensation*, *dew point*.

REMNANT ARC A formerly active volcanic island arc that has been moved away